

Impinj xArray Gateway

The Impinj xArray Gateway is a fixed infrastructure RAIN RFID system providing always-on, wide-area monitoring, location, and direction of items. With a field of view up to 1,500 ft² (139 m²), integrated phased-array antenna with 52 beams in 9 sectors, and simplified PoE deployment, it is designed for retail, healthcare, and manufacturing item-level applications.



UHF RFID

Operating Frequencies	FCC: 902-928 MHz; EU1: 865-868 MHz; EU2: 915-921 MHz; GX1/GX2/GX3; JP2: 916-921 MHz
Radiated Power (max)	FCC: 4W EIRP AC/PoE; EU1: 2W ERP AC/PoE; EU2: 4W ERP AC/PoE+
Operating Roles	3 (inventory, direction, location)
Antenna Beams	52
Maximum Read Rate	1100 tags/s
Antenna Type	Phased array with dual linear polarized antenna elements
3dB beam width (sum of all beams)	116° major axis (120° for EU1)
Coverage Area (typical)	1,500 ft ² / 139 m ² (2,000 ft ² / 186 m ² for EU2)
Air Interface Protocol	RAIN RFID: EPCglobal UHF Class 1 Gen2v2 / ISO 18000-63
Receive Sensitivity (max)	-84 dBm

MECHANICAL

Dimensions (H x W x D)	18.0 x 18.0 x 3.0 in (46.7 x 46.7 x 7.5 cm)
Weight	17.8 lb (8.0 kg)

CONNECTIVITY

Network Connectivity	10/100BASE-T
USB Ports	1 device, 1 host
Management Console Port (RS-232)	RJ-45
Power Sources	AC-DC power supply (all models); IEEE 802.3af PoE (except EU2); 802.3at PoE+ (EU2)

ENVIRONMENTAL

Operating Temperature	-4 °F to 122 °F (-20 °C to 50 °C)
Storage Temperature	-4 °F to 122 °F (-20 °C to 50 °C)
Humidity	95% non-condensing
Environmental Sealing (IEC)	IP50
RoHS Compliant	Yes
Reliability	Enterprise Grade

SOFTWARE

Network Protocols	SSH, HTTP, HTTPS, NTP, DHCP, SFTP, mDNS
Networking Stack	IPv4, IPv6
Management	SNMPv1, v2, v3
Security	802.1x (Port Security), TLS 1.2 for Secure LLRP
Host Interface	LLRP with Impinj Extensions
Custom Application Partition	Yes
Development Libraries	Octane SDK (.NET, Java), LTK (C, C++, .NET, Java), ETK (C, C++)

FEATURE HIGHLIGHTS

- Continuous, hands-free overhead monitoring of a wide area
- Field of view up to 1,500 ft² (139 m²) per single gateway
- Real-time item identification, location, and movement on two axes
- Resolves x-y locations to within 5 ft (1.5 m)
- Integrated phased array antenna with 52 dual-linear polarized beams in 9 sectors
- Power and control through a single PoE connection

PRODUCT GROUP

Category	RFID Reader
Product Group	Fixed RAIN RFID Gateway